

Equipment Maintenance & Inspection Record

Document ID: MR-2026-0114 | Facility: Plant 3 - Utilities Block | Confidential - Internal Use

1. Equipment Details

Equipment Tag	P-204
Description	Centrifugal Transfer Pump, Cooling Water Circuit
Location	Utilities Block, Bay 3
Manufacturer / Model	Kirloskar KDS 150-315
Installed On	March 2019
Criticality Rating	High (linked to cooling water supply for Coke Oven Battery 2)

2. Maintenance History Log

Date	Work Order	Description	Technician	Status
12-Feb-2026	WO-4471	Routine bearing lubrication and vibration check	R. Sharma	Closed
03-Apr-2026	WO-4528	Replaced mechanical seal - minor leakage observed	A. Verma	Closed
19-May-2026	WO-4602	Abnormal vibration reported by shift operator during routine check	R. Sharma	Closed
21-May-2026	WO-4605	Coupling alignment corrected; vibration reading normal	S. Jeyaraj	Closed
30-Jun-2026	WO-4671	Quarterly preventive maintenance - full inspection	A. Verma	Closed

3. Inspection Findings (30-Jun-2026 Quarterly PM)

During the quarterly preventive maintenance inspection, the following observations were recorded: the pump casing showed minor surface corrosion near the discharge flange, attributed to prolonged exposure to humid conditions in Bay 3. No active leakage was found at the mechanical seal following the April repair. Vibration readings were within normal operating range (2.1 mm/s RMS, baseline 1.8-2.3 mm/s). Motor winding insulation resistance tested satisfactory. Recommendation: schedule casing surface treatment during the next planned shutdown window; no immediate operational risk identified.

Cross-reference: The May 2026 abnormal vibration event (WO-4602) was traced to coupling misalignment rather than bearing wear, corrected under WO-4605. No recurrence observed as of this inspection.

4. Applicable Safety Procedure Reference

Any maintenance activity on P-204 requiring isolation of the cooling water line must follow Permit-to-Work Procedure SOP-SAF-011 (Confined Space and Line Break Isolation). A hot work permit is required if any welding or grinding is performed within 5 metres of the pump due to proximity to the hydrocarbon vent line on the adjacent unit. Gas testing (LEL and H2S) must be completed and logged before work commences, and re-tested every 2 hours during the work window.

5. Regulatory Compliance Note

This equipment falls under the periodic inspection requirements of the Factory Act statutory guidelines for pressure-bearing rotating equipment. Last statutory inspection was completed on 15-Jan-2026 (Certificate No. FA-INSP-2026-0092, valid until 15-Jan-2027). No non-conformances were raised during the statutory

inspection.

Prepared by: A. Verma, Senior Maintenance Engineer | Reviewed by: Plant Safety Officer | Next scheduled PM: 30-Sep-2026